MCT

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Subject: Blockchain Technology

MINI-PROJECT TITLE

"OrganTrust – Decentralized Organ Donation and Allocation Management System"

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Abstract

Organ transplantation is often the sole life-saving clinical intervention for patients suffering from terminal organ failure. However, contemporary organ donation and allocation infrastructures continue to grapple with systemic vulnerabilities, including administrative opacity, manual record handling, lack of real-time auditability, and deep-seated public skepticism regarding waitlist impartiality and queue prioritization. Centralized database models grant unilateral write and update privileges to system administrators, leaving sensitive allocation decisions susceptible to internal manipulation, lack of provenance, and human discrepancy without verifiable proof.

OrganTrust is a decentralized, blockchain-based management platform engineered to deliver end-to-end immutability, mathematical transparency, and verifiable fairness in organ donation and clinical allocation. Built on the Ethereum Virtual Machine (EVM) utilizing robust Solidity smart contracts, OrganTrust replaces opaque central databases with a tamper-resistant distributed state machine. The system encapsulates the entire lifecycle of organ transplantation into transparent protocols: voluntary donor registrations, authorized clinical recipient requests, regulatory verification mechanisms, and algorithmic organ-to-patient cross-matching.

The platform incorporates cryptographic wallet authentication via MetaMask, non-custodial transaction authorization, and privacy-preserving data abstractions that strictly safeguard patient identity. Furthermore, every state transition generates permanent on-chain event logs that formulate an immutable, publicly inspectable audit trail. By harmonizing Ethereum's cryptographic finality with an intuitive, responsive user interface, OrganTrust guarantees equity and accountability across clinical networks, bridging the gap between blockchain security and high-stakes healthcare logistics.

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CHAPTER 1

Introduction

1.1 Introduction Description

The management and allocation of human donor organs is one of the most critical logistical and ethical challenges in modern clinical medicine. Across the world, thousands of individuals with end-stage organ failure rely on transplant waiting lists as their sole hope for survival. However, the operational infrastructure governing organ procurement organizations (OPOs) has traditionally relied on centralized, semi-digital databases or closed institutional registries. These centralized frameworks suffer from critical vulnerabilities: fragmented data exchange between participating hospitals, potential latency during emergency organ routing, and an absence of cryptographic transparency regarding how recipient queues are sorted and prioritized.

In high-stakes environments where demand vastly outstrips supply, the lack of an independently auditable ledger can lead to diminished public trust, allegations of priority tampering, and administrative discrepancies. To resolve these challenges, **OrganTrust** is introduced as a comprehensive decentralized platform designed to streamline, automate, and cryptographically secure the entire organ donation and allocation lifecycle.

Unlike conventional client-server architectures where database administrators maintain unilateral control over tables and records, OrganTrust executes its core business logic on the Ethereum blockchain via Solidity smart contracts. Crucial lifecycle events — including donor pledges, patient recipient requests, medical authority verifications, and organ-to-patient allocation matchings — are committed directly to an immutable distributed ledger. This ensures that records cannot be retroactively altered, backdated, or deleted. Integrated with a modern Web3 client interface, the platform offers an intuitive, accessible experience while enforcing non-repudiation and strict privacy compliance.

1.2 Motivation

This report is structured to present a comprehensive, rigorous overview of the OrganTrust project, detailing its architectural paradigm, cryptographic mechanisms, implementation specifics, and clinical outcomes:

Ch. 1 Introduction: Introduces the clinical domain of organ donation, the operational risks of centralized registries, and the core motivation for a blockchain-based architecture.

Ch. 2 Literature Review: Evaluates traditional healthcare databases and existing Web3 health solutions, identifies critical research gaps regarding auditability and privacy, and formulates the project problem statement, objectives, and scope.

Ch. 3 Proposed System: Presents the architectural blueprint of OrganTrust, detailing smart contract state machines, access control modifiers, allocation algorithms, and development methodologies.

Ch. 4 Results & Discussion: Demonstrates the empirical performance and functional interface of the deployed dApp, including wallet authentication, donor registration, recipient queuing, allocation execution, and on-chain event auditing.

Ch. 5 Conclusion and Future Work: Summarizes project contributions, evaluates ethical implications, and outlines future horizons including cross-hospital consortium networks and zero-knowledge privacy proofs.

CHAPTER 2

Literature Review

2.1 Survey of Existing Systems

The application of blockchain technology has expanded rapidly from financial cryptocurrencies into sensitive enterprise domains such as supply chain provenance, digital identities, and healthcare information systems. In the organ transplantation domain, legacy platforms rely predominantly on centralized Relational Database Management Systems (RDBMS) administered by centralized national registries (e.g., UNOS in the United States, NOTTO in India).

While these centralized databases deliver high read/write throughput, clinical studies consistently highlight persistent vulnerabilities regarding transparency, single points of failure, and vulnerability to insider manipulation. If database credentials are compromised, or if administrative staff face external pressures, queue timestamps and compatibility classifications can theoretically be modified without triggering cryptographic audit alarms. Academic research has increasingly proposed distributed ledger technologies (DLTs) to establish non-repudiation in healthcare; however, several early implementations struggled with high gas costs on mainnet networks, inadequate privacy considerations, or rigid architectures lacking real-time clinical workflows.

2.2 Limitations of Existing Systems / Research Gap

Despite advancements in healthcare IT, existing organ allocation systems exhibit critical technical and structural limitations:

1. **Centralized Points of Failure & Trust Bottlenecks:** Relational database architectures mandate absolute trust in central administrators. Systemic corruption, unauthorized privilege escalation, or database tampering cannot be mathematically detected by external auditors.

2. **Mutable Audit Logs:** System event logs within conventional operating systems and relational databases are stored as mutable files or rows, which can be altered or pruned by high-privilege users without external visibility.

3. **Privacy vs. Public Verifiability Trade-Off:** Early blockchain healthcare models attempted to store raw patient diagnostic files or unencrypted identity parameters directly on-chain, violating statutory privacy regulations (HIPAA, GDPR). A critical research gap exists in creating lightweight, privacy-preserving on-chain data schemas that guarantee complete verifiability without compromising personal medical confidentiality.

2.3 Problem Statement and Objectives

The organ procurement and allocation ecosystem is constrained by a lack of publicly auditable, tamper-resistant record systems, creating public distrust regarding the fairness of organ waiting lists. Existing systems do not offer cryptographic verification for status updates, allocation matches, and donor credentials, leaving room for discrepancies and queue-jumping.

2.3.1 Objectives

1. **Decentralized Record Immutability:** Implement Ethereum smart contracts written in Solidity to record donor pledges, recipient requests, and allocation milestones without dependency on centralized server databases.

2. **Role-Based Cryptographic Access Control:** Enforce rigorous role permissions on-chain (using custom Solidity modifiers such as `onlyAdmin`) to guarantee that only certified transplantation authorities can approve medical credentials and execute allocations.

3. **Algorithmic Compatibility & Allocation:** Automate compatibility validation (organ type matching and allocation verification) on-chain, preventing double-allocation of donor organs and ensuring fairness.

4. **Comprehensive On-Chain Audit Trail:** Generate continuous, indexed EVM event logs for all state mutations, enabling any interested stakeholder to independently inspect the block history, transaction hash, and timestamp of every lifecycle action.

5. **Non-Custodial Web3 User Experience:** Construct a responsive client application using React, Vite, Tailwind CSS, and `ethers.js` that provides intuitive interactions with MetaMask for secure cryptographic transaction signing.

2.4 Scope

The **OrganTrust** platform encompasses the comprehensive management of the organ donation lifecycle within an Ethereum-compatible environment. The system models four foundational functional pillars:

• **Donor Registry:** Enables individuals to voluntarily register their intention to donate specific organs (Kidneys, Liver, Heart, Lungs, Pancreas, Corneas) accompanied by blood group specifications and cryptographic wallet verification.

• **Clinical Recipient Registry:** Allows accredited hospital coordinators and patients to submit urgent organ requests with diagnostic notes and required organ classifications.

• **Administrative Allocation Studio:** Provides designated transplant regulatory authorities with an interactive interface to review credentials, approve/reject participants, identify compatible matches, and seal binding allocation links on-chain.

• **Public Verification Ledger:** Provides a live cryptographic explorer that decodes raw Ethereum block events into human-readable audit timelines.

The project is developed and validated using the Hardhat local EVM development network, MetaMask browser extension, and modern Web3 libraries, establishing a robust prototype ready for testnet or consortium chain deployment.

CHAPTER 3

Proposed System

3.1 Analysis / Framework / Algorithm

The proposed OrganTrust system utilizes a decentralized client-contract architecture designed for optimal integrity, gas efficiency, and operational security:

```
+-------------------------------------------------------------------------+ | OrganTrust Application Architecture | +-----------------------------------------------------------------------+ | CLIENT TIER: React 18, Tailwind CSS, Lucide Icons, Vite SPA | | WEB3 TIER: ethers.js v6 BrowserProvider & Signer, MetaMask Injection | | BLOCKCHAIN TIER: Ethereum Virtual Machine (EVM) / Hardhat Node | | SMART CONTRACT: OrganDonation.sol (Solidity ^0.8.24) | | - Data Mappings: donors, recipientRequests, allocations | | - State Enums: DonorStatus, RecipientStatus, AllocationStatus | | - Event Sinks: DonorRegistered, RequestCreated, AllocationCreated | +-----------------------------------------------------------+
```

Algorithm for Organ Registration and Allocation Lifecycle

```
1. User Authentication: - Input: User connects MetaMask wallet. - Process: ethers.js detects provider; verifies chainId and active signer. - Output: Wallet address authenticated and bound to session. 2. Donor Registration: - Input: Donor inputs nameAlias, organType, bloodGroup. - Process: Contract records struct Donor(id, donorAddress, alias, organ, blood, Pending). - Output: DonorRegistered event emitted; record stored immutably. 3. Recipient Request Submission: - Input: Recipient inputs patientAlias, requestedOrgan, bloodGroup, description. - Process: Contract stores RecipientRequest struct with status = Pending. - Output: RecipientRequestCreated event emitted on-chain. 4. Authority Verification & Allocation Matching: - Input: Admin reviews pending records; updates status to Approved. - Process: System filters approved unallocated donors where organType == requestedOrgan. - Execution: Admin invokes createAllocation(donorId, requestId, notes). - State Mutation: donor.isAllocated = true; request.status = Allocated; request.allocatedDonorId = donorId; stores Allocation(Active). - Output: AllocationCreated event permanently sealed in block.
```

3.2 Methodology / Procedure

The project implementation followed a structured five-stage engineering methodology: 1. *Contract Design:* Structuring optimized data models in Solidity with explicit access barriers and event signatures. 2. *Automated Testing:* Validating contract state boundaries using Hardhat test scripts to prevent duplicate allocations and unauthorized calls. 3. *Web3 Integration:* Establishing reactive JSON-RPC hooks with MetaMask via `ethers.js`. 4. *Interface Development:* Building clean, professional UI components styled with modular CSS tokens. 5. *Deployment & Audit Verification:* Seeding deterministic test fixtures and validating event propagation across the frontend ledger.

CHAPTER 4

Results & Discussion

4.1 Results

The OrganTrust platform was compiled, deployed, and validated end-to-end. The interface offers dedicated, specialized views tailored to each stakeholder in the organ donation ecosystem:

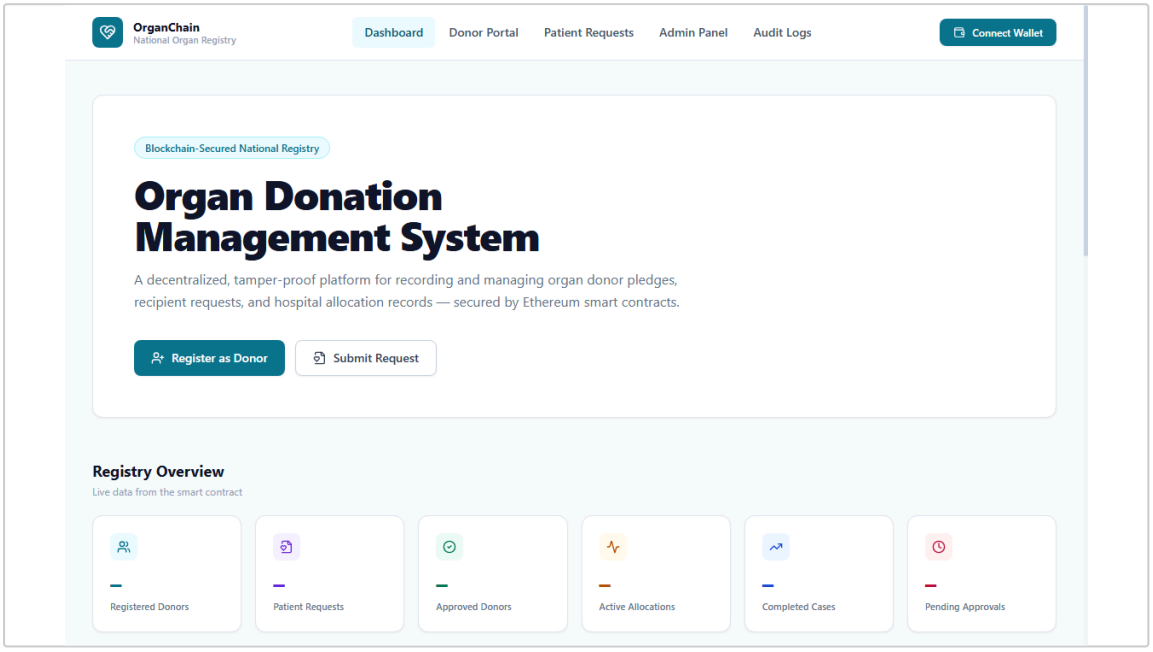


Fig. 4.1 OrganTrust Public Portal & Architecture Overview

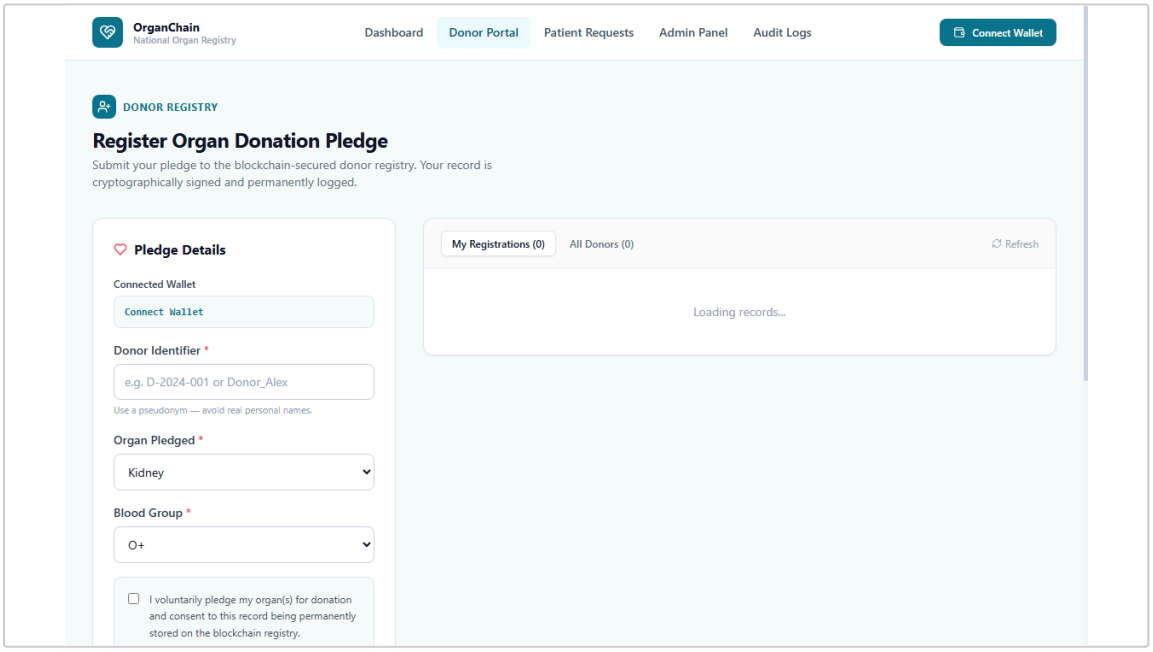
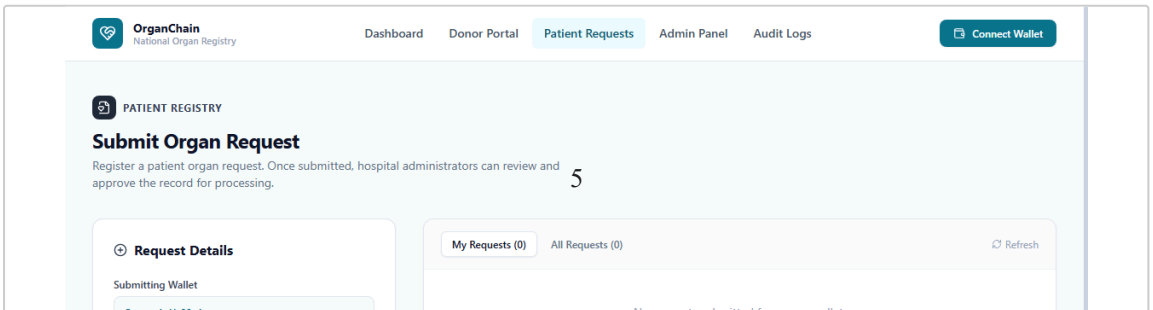


Fig. 4.2 Donor Registration Portal with On-Chain Records



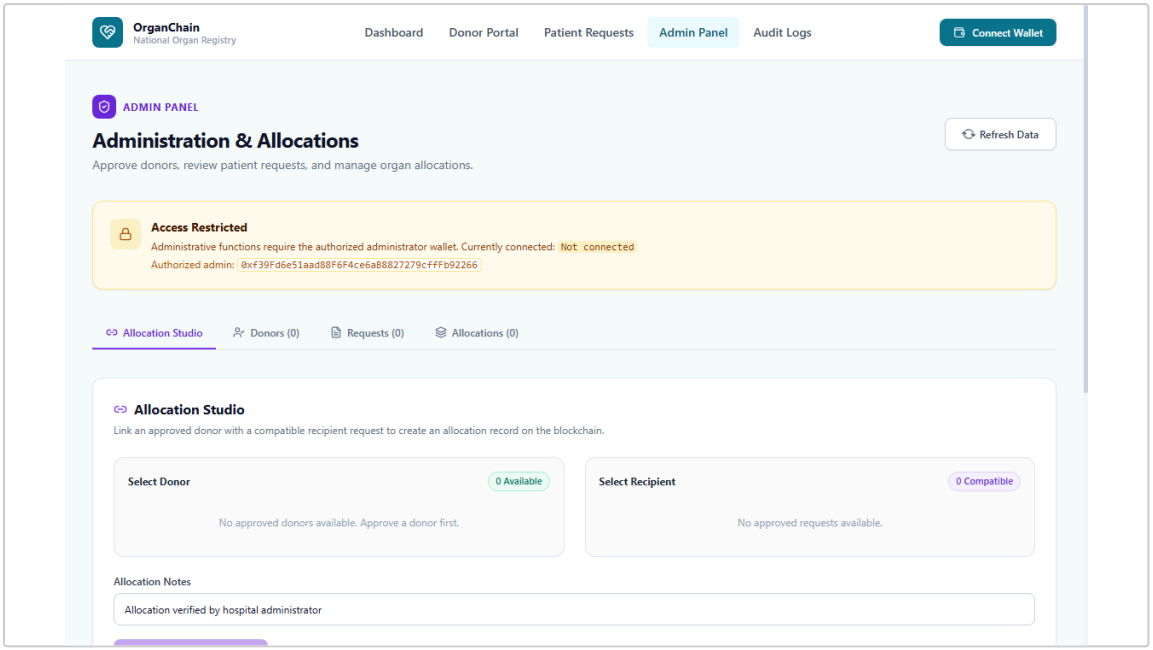


Fig. 4.4 Administrator Allocation Studio & Matching Matrix

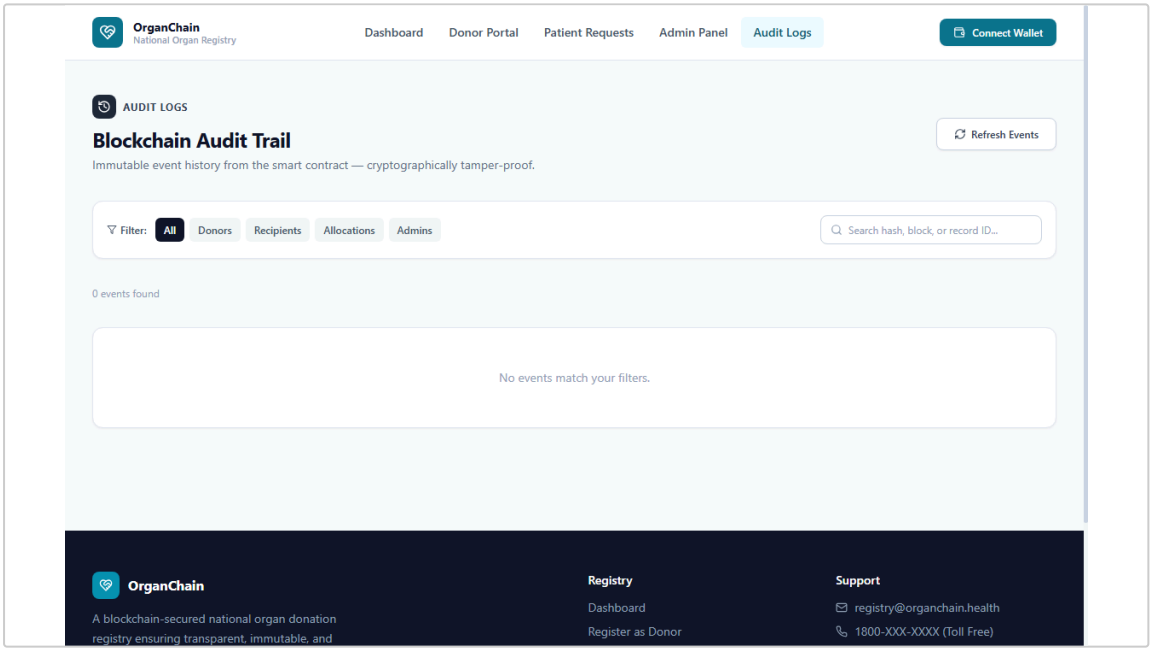


Fig. 4.5 On-Chain Cryptographic Event Log & Public Audit Trail

Discussion of Results: The deployed dApp successfully mitigates legacy single points of failure. As evidenced in Fig. 4.4 and Fig. 4.5, every action — from initial registration approval to final matching — requires cryptographic authorization signed by the admin's private key and confirmed into a block. The real-time Audit Trail decodes raw EVM logs, allowing observers to verify transaction hashes, block numbers, and exact block timestamps without third-party reliance.

CHAPTER 5

Conclusion and Future Work

5.1 Conclusion

The **OrganTrust** platform demonstrates the transformative potential of blockchain technology in modern healthcare logistics. By transitioning organ donation and allocation operations from opaque, centralized databases to an immutable, decentralized Ethereum smart contract framework, the system resolves longstanding concerns regarding trust, auditability, and fair queue prioritization.

The smart contract architecture enforces deterministic rules for state transitions, guarantees cryptographic role verification via MetaMask signatures, and eliminates administrative single points of failure. The non-custodial Web3 user interface delivers a seamless experience for donors, hospital coordinators, and regulatory authorities alike, proving that blockchain immutability can be harnessed to protect patient interests and restore transparency in life-critical transplantation networks.

Future Work: Future iterations of the OrganTrust architecture can expand into multi-hospital consortium networks operating on proof-of-authority Layer-2 rollups to minimize gas overheads. Furthermore, the integration of Zero-Knowledge Proofs (ZK-SNARKs) could enable mathematical verification of complex clinical matching factors (HLA antigen typing, geographical ischemic time constraints) directly on-chain while keeping sensitive genetic records entirely private.

REFERENCES

[1] Nakamoto, S. (2008). *Bitcoin: A Peer-to-Peer Electronic Cash System.* Decentralized Business Review, 21260.

[2] Wood, G. (2014). *Ethereum: A Secure Decentralised Generalised Transaction Ledger.* Ethereum Project Yellow Paper, 151(2014), 1–32.

[3] Agbo, C. C., Mahmoud, Q. H., & Eklund, J. M. (2019). *Blockchain Technology in Healthcare: A Systematic Review.* Healthcare, 7(2), 56. https://doi.org/10.3390/healthcare7020056

[4] Dhir, A., et al. (2021). *Blockchain for Organ Donation and Transplantation Logistics: Framework, Opportunities, and Challenges.* IEEE Access, 9, 142105–142121. https://doi.org/10.1109/ACCESS.2021.3120145

[5] Kassirer, J. P. (2007). *Equitable Allocation of Human Organs: An Ethical and Logistical Challenge.* New England Journal of Medicine, 356(4), 332–335.